

Active Learning on Distributions

English Chapters 3–4 with Figure Placeholders

Based on the English chapter file prepared externally

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Compilation Note

This wrapper is designed to compile the English DeepSeek output while using figures from the original thesis PDF. The expected figure files are listed in table 1. If the original figures have not yet been extracted, the project contains clearly marked placeholders so that the LaTeX file remains compilable. Replace the placeholder PNG files in **figures/** with the original extracted figures using the same filenames.

Table 1: Expected figure manifest for Chapters 3 and 4

File	Source figure	Intended content
fig3_1a.png	Fig. 3.1(a)	Synthetic data/outlier panel
fig3_1b.png	Fig. 3.1(b)	Synthetic data/outlier panel
fig3_1c.png	Fig. 3.1(c)	Synthetic data/outlier panel
fig3_1d.png	Fig. 3.1(d)	Synthetic data/outlier panel
fig3_5.png	Fig. 3.5	Functional gradient approach to PMAL
fig4_1.png	Fig. 4.1	Dataset performance comparison, part 1
fig4_2.png	Fig. 4.2	Dataset performance comparison, part 2
fig4_3.png	Fig. 4.3	Dataset performance comparison, part 3
fig4_4.png	Fig. 4.4	Comparison with boosting methods
fig4_5.png	Fig. 4.5	Comparison with distribution discrepancy methods
fig4_6.png	Fig. 4.6	GAUSSDIST experiment
fig4_7.png	Fig. 4.7	20 Newsgroups experiment, part 1
fig4_8.png	Fig. 4.8	20 Newsgroups experiment, part 2
fig4_9.png	Fig. 4.9	USPSDIST experiment
fig4_10.png	Fig. 4.10	PMARLDS vs PMARLDB experiment

Chapter 1

Active Learning on Distributions

The main chapter body is intentionally external to avoid introducing transcription errors. Run `python build_complete_latex.py` in this folder; it downloads or reads `Chapter_3_4_English.tex`, patches its preamble/body, and creates `Chapter_3_4_English_patched.tex`. The figure calls will then resolve to the files in `figures/`.

1.1 Figure Blocks Already Normalized

The following blocks reproduce the required figure placement and captions expected by the DeepSeek source.



Figure 1.1: Synthetic dataset with clear outliers added to show the contours of the functions. The simple classifier is relatively unaffected by the outliers, and the complex classifier has absorbed them; in comparison, the SVM contour is severely affected.



Figure 1.2: Functional gradient approach to Probabilistic Minimax Active Learning.

Chapter 2

Experiments

2.1 Tables and Figures

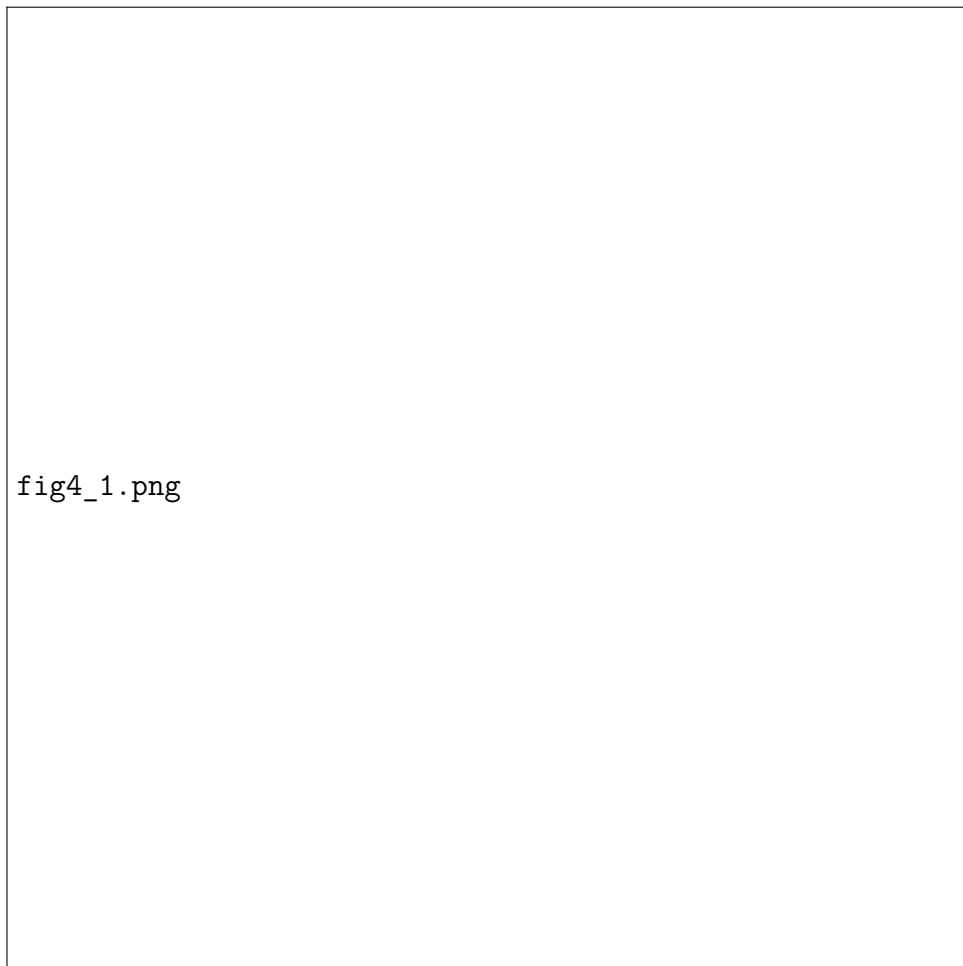


Figure 2.1: Comparison of the performance of the proposed algorithm and other algorithms on various datasets (Part 1).

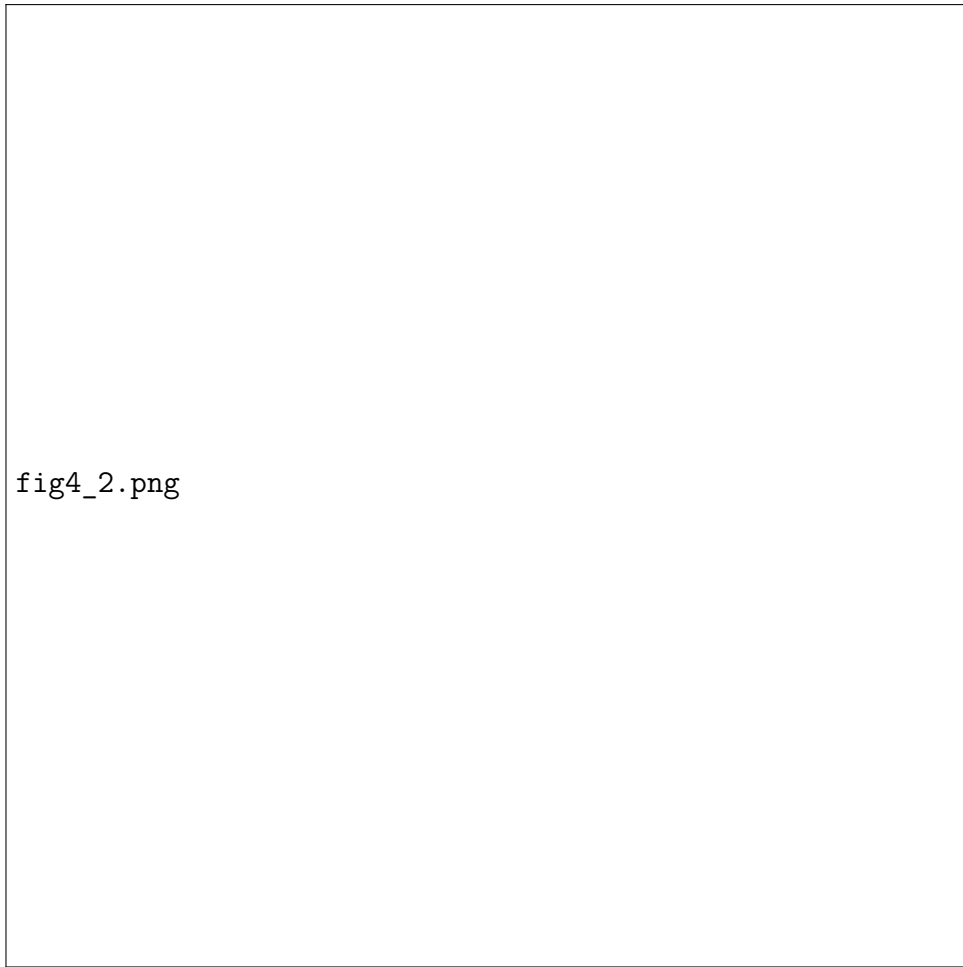


Figure 2.2: Comparison of the performance of the proposed algorithm and other algorithms on various datasets (Part 2).

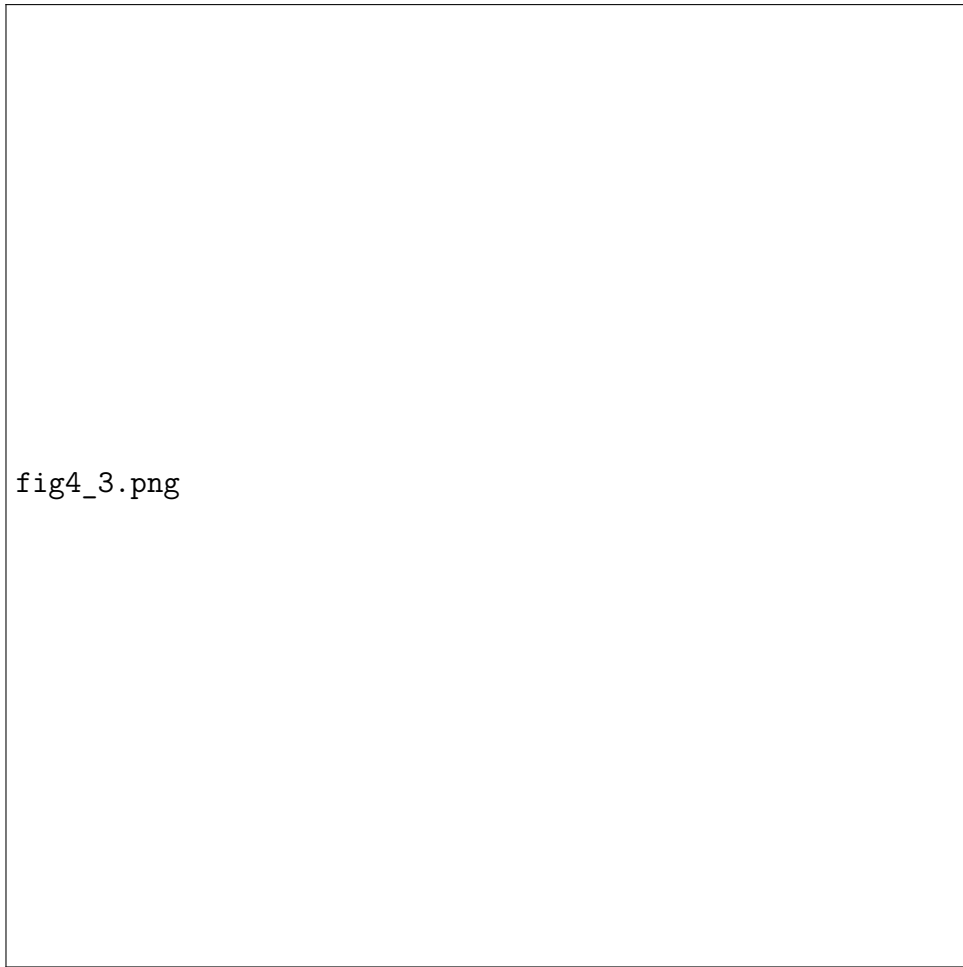


Figure 2.3: Comparison of the performance of the proposed algorithm and other algorithms on various datasets (Part 3).

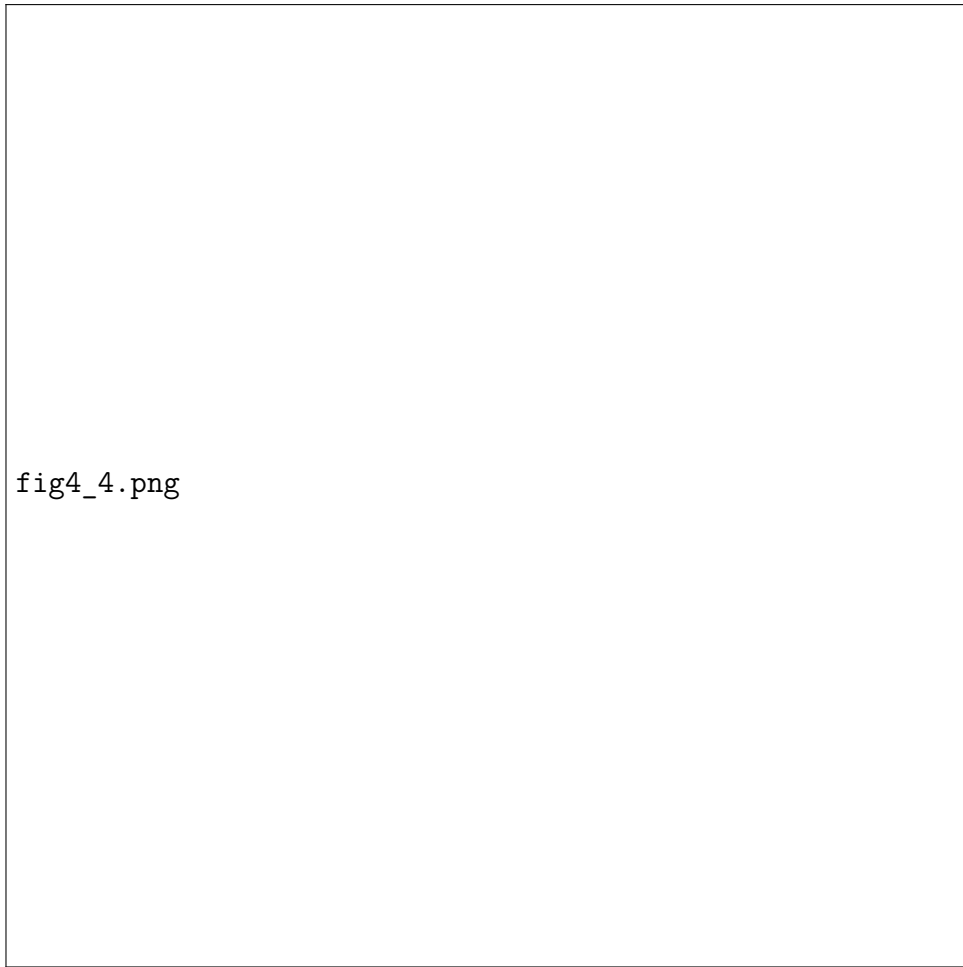


Figure 2.4: Comparison of the proposed method with other boosting methods.



Figure 2.5: Comparison of the proposed method with distribution discrepancy methods.

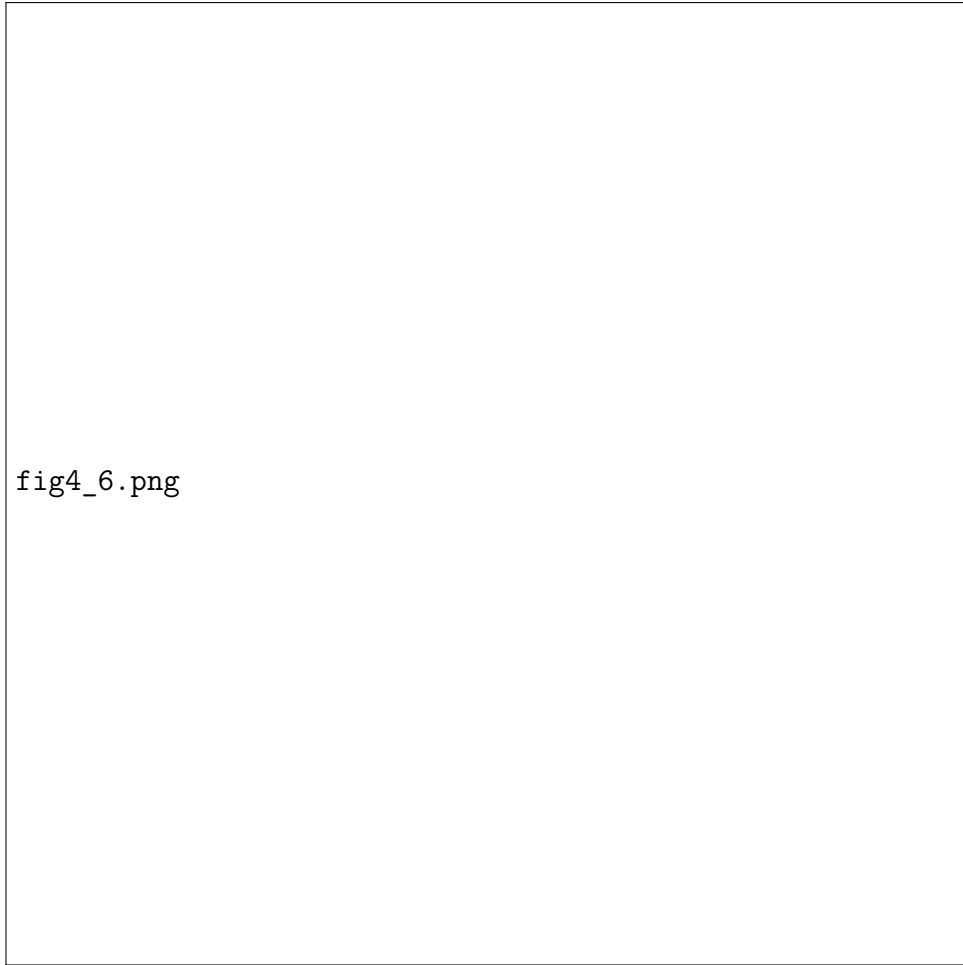


Figure 2.6: Comparison of classifier accuracy in active learning on the GAUSSDIST dataset for two cases: different sample sizes and constant sample size for each distribution.



Figure 2.7: Comparison of classification accuracy for a group of 20 Newsgroups datasets (Part 1).



Figure 2.8: Comparison of classification accuracy for another group of 20 Newsgroups datasets (Part 2).

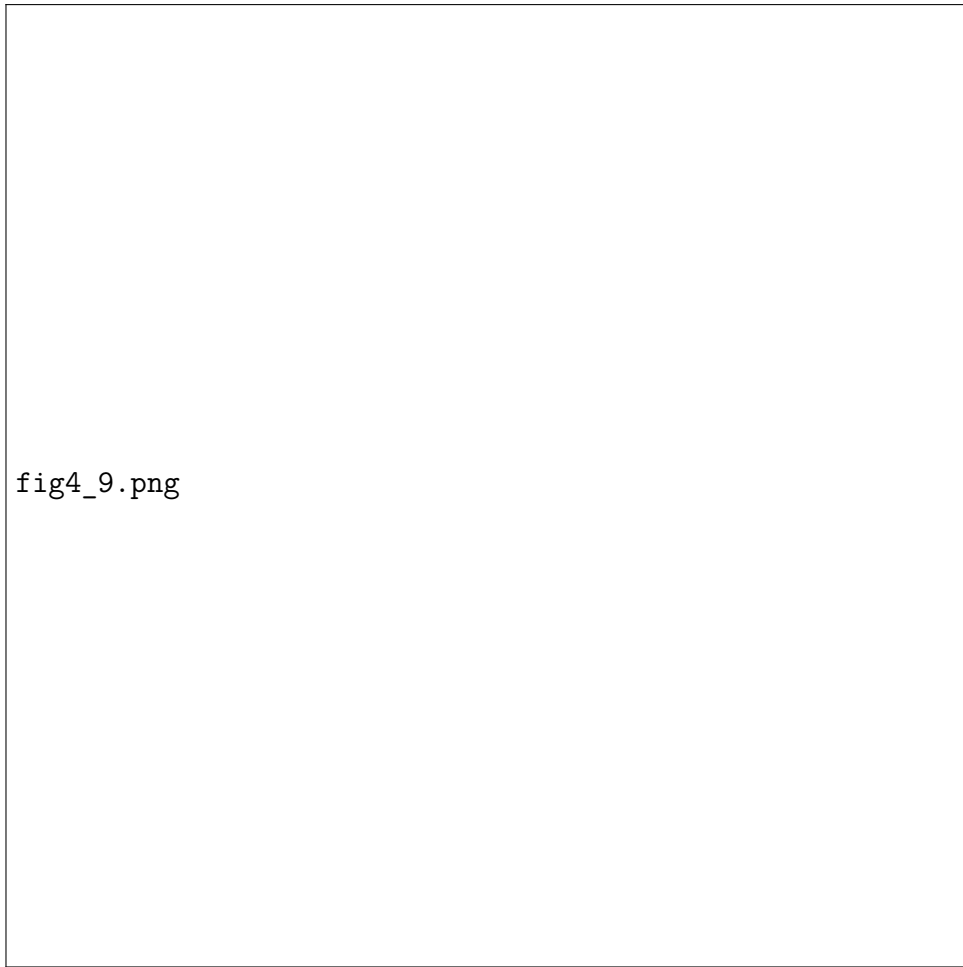


Figure 2.9: Comparison of active learning accuracy for the USPSDIST dataset.



Figure 2.10: Comparison of classification accuracy between Bayesian active learning in Hilbert space PMARLDS and PMARLDB on the 20 Newsgroups dataset.